

Pontryagin-Based Solver with Universally Smoothed Hamiltonian, Adaptive ΔT , and PA-Bundle Refinement

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① 1D Example

Formulation

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Example

Let $B = [-1, 1]$, $f = \alpha$, $h = x^2/2$ and $g = 0$, then our optimal control problem becomes

$$\begin{aligned} X'(t) &= \alpha(t), \quad 0 < t < T \\ X(0) &= X_0 \end{aligned} \tag{1}$$

$$u(X_0, 0) = \inf_{\alpha \in [-1, 1]} \int_0^T \frac{X^2(s)}{2} ds \tag{2}$$

Introduce the regularized paths

$$\begin{aligned} \bar{X}_{n+1} &= \bar{X}_n + \Delta t H_\lambda^\delta(\bar{\lambda}_{n+1}, \bar{X}_n), \quad \bar{X}_0 = X_0, \\ \bar{\lambda}_n &= \bar{\lambda}_{n+1} + \Delta t H_x^\delta(\bar{\lambda}_{n+1}, \bar{X}_n), \quad \bar{\lambda}_N = g_x(\bar{X}_N) \end{aligned} \tag{3}$$

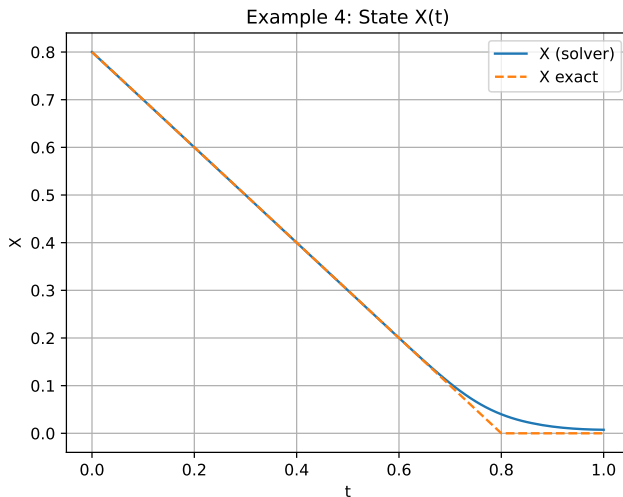


Figure: Solution $\bar{X}$ of the discrete optimization problem (1)-(2) with $X_0 = 0.8$

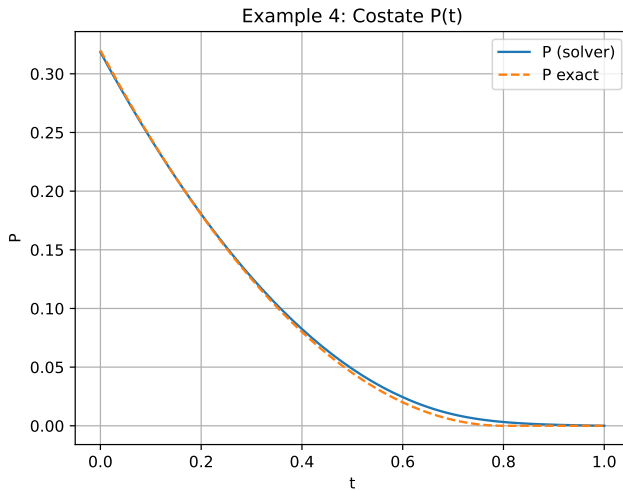


Figure: Solution $\bar{\lambda}$ of the discrete optimization problem (1)-(2) with $\lambda_T = 0$

Parameters

- $N = 51$ nodes, $T = 1$, $dt = 0.02$
- $\varepsilon_{\text{time}} = 1e-3$, $\varepsilon_{\text{PA}} = 1e-4$, $\varepsilon_{\delta} = 2e-3$
- $\text{maxiter} = 8$

Results

- $N_{\text{final}} = 64$
- $\delta = 0.005$, $M = 3$
- $\eta_{\text{time}} = 0.00096718$, $\eta_{\text{PA}} = 0.0$, $\eta_{\delta} = 0.00110596$
- newton residual = $4.2749e-13$

From Sandberg references: Symplectic Euler method

$$\begin{aligned}\bar{X}_{n+1} &= \bar{X}_n + \Delta t H_\lambda(\bar{\lambda}_{n+1}, \bar{X}_n), & \bar{X}_0 &= X_0, \\ \bar{\lambda}_n &= \bar{\lambda}_{n+1} + \Delta t H_x(\bar{\lambda}_{n+1}, \bar{X}_n), & \bar{\lambda}_N &= g_x(\bar{X}_N)\end{aligned}\tag{4}$$

Implementation:

$$\begin{aligned}\bar{X}_{n+1} &= \bar{X}_n + \Delta t H_\lambda^\delta(\bar{\lambda}_n, \bar{X}_n, t_n), & \bar{X}_0 &= X_0, \\ \bar{\lambda}_n &= \bar{\lambda}_{n+1} + \Delta t H_x^\delta(\bar{\lambda}_{n+1}, \bar{X}_{n+1}, t_{n+1}), & \bar{\lambda}_N &= g_x(\bar{X}_N)\end{aligned}\tag{5}$$

Sparsity Pattern Residual

We consider the case $N = 3$. The values X_0 and $\bar{\lambda}_3 = g_x(\bar{X}_3) = 0$ are given and our system has the variables

$$z = (\bar{\lambda}_0, \bar{X}_1, \bar{\lambda}_1, \bar{X}_2, \bar{\lambda}_2, \bar{X}_3)^\top \in \mathbb{R}^6. \quad (6)$$

with

$$F(z) = \begin{pmatrix} \bar{\lambda}_0 - \bar{\lambda}_1 - \Delta t \bar{X}_0 \\ \bar{X}_1 - \bar{X}_0 + \Delta t \tanh(\bar{\lambda}_1/\delta) \\ \bar{\lambda}_1 - \bar{\lambda}_2 - \Delta t \bar{X}_1 \\ \bar{X}_2 - \bar{X}_1 + \Delta t \tanh(\bar{\lambda}_2/\delta) \\ \bar{\lambda}_2 - \bar{\lambda}_3 - \Delta t \bar{X}_2 \\ \bar{X}_3 - \bar{X}_2 + \Delta t \tanh(\bar{\lambda}_3/\delta) \end{pmatrix}. \quad (7)$$

Thus we obtain the following Newton Iteration system

$$\begin{pmatrix} 1 & -1 & & & & \\ & d_1^i \Delta t & 1 & & & \\ & 1 & -\Delta t & -1 & & \\ & & -1 & d_2^i \Delta t & 1 & \\ & & & 1 & -\Delta t & \\ & & & & -1 & 1 \end{pmatrix} \begin{pmatrix} \Delta \bar{\lambda}_0^i \\ \Delta \bar{\lambda}_1^i \\ \Delta \bar{X}_1^i \\ \Delta \bar{\lambda}_2^i \\ \Delta \bar{X}_2^i \\ \Delta \bar{X}_3^i \end{pmatrix} = \begin{pmatrix} \bar{\lambda}_0^i - \bar{\lambda}_1^i - \Delta t \bar{X}_0^i \\ \bar{X}_1^i - \bar{X}_0^i + \Delta t \tanh(\bar{\lambda}_1^i/\delta) \\ \bar{\lambda}_1^i - \bar{\lambda}_2^i - \Delta t \bar{X}_1^i \\ \bar{X}_2^i - \bar{X}_1^i + \Delta t \tanh(\bar{\lambda}_2^i/\delta) \\ \bar{\lambda}_2^i - \bar{\lambda}_3^i - \Delta t \bar{X}_2^i \\ \bar{X}_3^i - \bar{X}_2^i + \Delta t \tanh(\bar{\lambda}_3^i/\delta) \end{pmatrix} \quad (8)$$

and $d_j^i := \partial_{\lambda} \tanh(\bar{\lambda}_j^i/\delta) = \delta^{-1} \cosh^{-2}(\bar{\lambda}_j^i/\delta)$, where $d_j^i \Delta t = \cosh^{-2}(\bar{\lambda}_j^i/\delta)$ since we are taking $\delta = \Delta t$.

$$z = (\bar{X}_1, \bar{X}_2, \bar{X}_3, \bar{\lambda}_0, \bar{\lambda}_1, \bar{\lambda}_2)^\top \quad (9)$$

Jacobian pattern (current z-order: x-block then p-block)

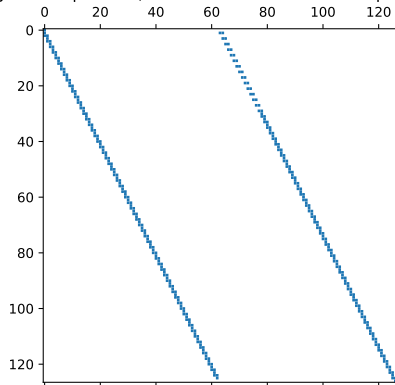


Figure:

$$z = (\bar{\lambda}_0, \bar{X}_1, \bar{\lambda}_1, \bar{X}_2, \bar{\lambda}_2, \bar{X}_3)^T \in \mathbb{R}^6. \quad (10)$$

Jacobian pattern after COLUMN interleaving (time-ordered unknowns)

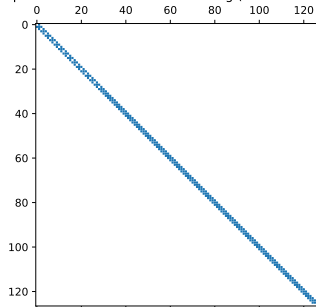


Figure:



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